$v_O = 0$, $O \equiv CM$, $v_{CM} = 0$

$$\boxed{\bar{M}_O^E = \frac{d\bar{L}_O}{dt}}$$

$$\bar{F} = m\bar{a} = \frac{d\bar{p}}{dt}$$

$$\bar{M} = \bar{r} \times \bar{F}$$

$$\bar{L} = \bar{r} \times \bar{p}$$

$\bar{\omega} = \omega \bar{u}_z$

z asse p. i.

$$\bar{L}_O = \bar{\bar{I}}\,\bar{\omega} = I_{zz}\bar{\omega} + \bar{L}_\perp = I_z \bar{\omega}$$

$$\bar{M}_O^E = \frac{d\bar{L}_O}{dt} = \frac{d}{dt}\left(I_z \bar{\omega}\right) = I_z \frac{d\bar{\omega}}{dt} = I_z \bar{\alpha}$$

$$\boxed{\bar{M}_O^E = I_z \bar{\alpha}}$$

← eq. differenziale del moto di rotazione

$$\boxed{\bar{\alpha} = \frac{\bar{M}_O^E}{I_z}} \Rightarrow \boxed{\bar{\alpha} = \bar{\alpha}(t)}$$

$\alpha = \alpha(t) \Rightarrow$ moto di rotazione vario

$\alpha = \text{cost}\ (\neq 0) \Rightarrow$ moto di rotazione unif. accelerato

$$\Rightarrow \omega(t) = \omega_0 + \alpha t$$

$$\theta(t) = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$$

$\alpha = 0 \Rightarrow \omega(t) = \omega = \text{cost}$

$$\theta(t) = \theta_0 + \omega t$$

z non è asse principale di inerzia $L_\perp \neq 0$

$\bar{\omega} = \omega \bar{u}_z$

$$\Rightarrow \bar{L}_0 = I_z \bar{\omega} + \bar{L}_\perp$$

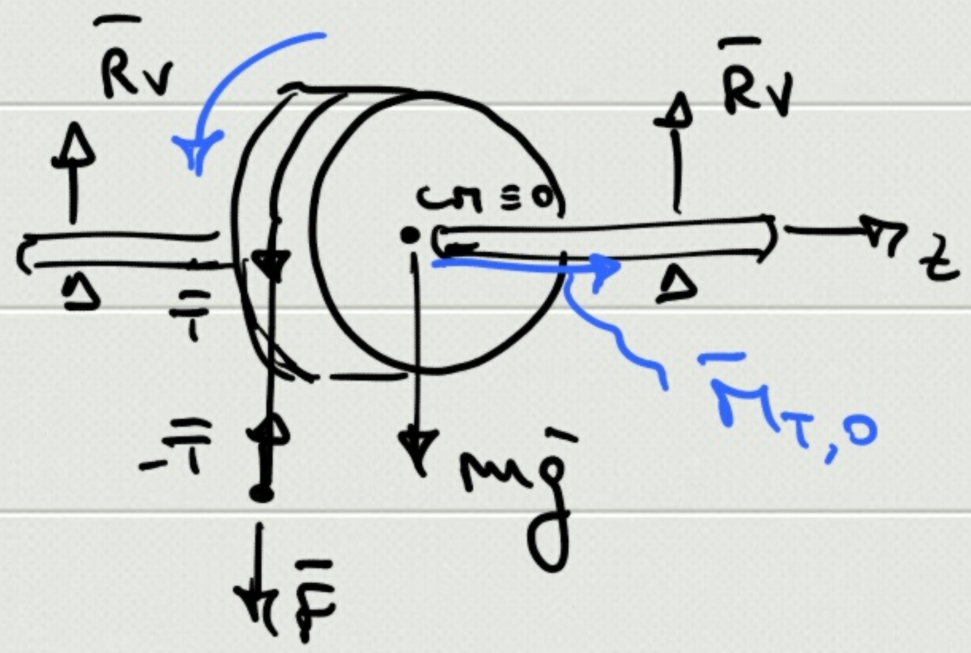
$$\bar{M}^E_0 = \frac{d\bar{L}_0}{dt} = \frac{d}{dt}\left(I_z \omega + \bar{L}_\perp\right) =$$

Variazione modulo $\bar{L}$

$$\Rightarrow \quad \bar{M}^{E}_{O} = I_z \bar{\alpha} + \frac{d\bar{L}_\perp}{dt}$$

variazione orientazione di $\bar{L}$

$$\bar{F} = \frac{d\bar{p}}{dt} = m \frac{dv}{dt} \bar{u}_T + m \frac{v^2}{R} \bar{u}_N$$



$m = 20$ kg

$R = 0.5$ m

$F = 9.8$ N

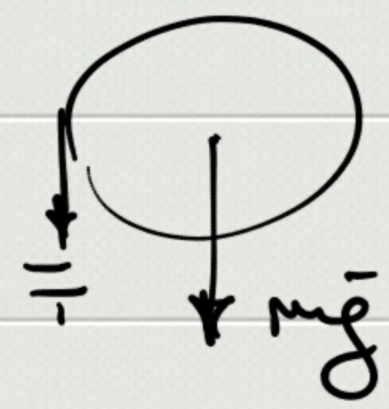
$\omega_0 = 0$

$\alpha = ?$ $\quad \omega(t = 2s) = ?$

CM ≡ polo

$v_{CM} = 0 \Rightarrow \bar{P} = 0 \Rightarrow \bar{R}^E = 0$

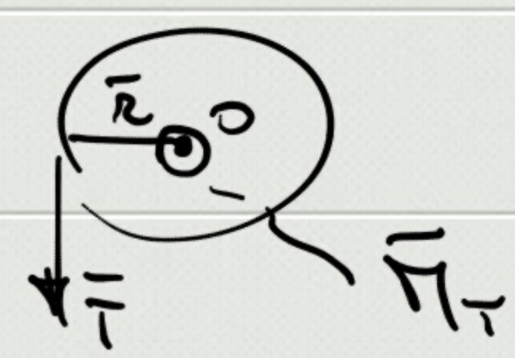
$$\overset{!}{=} m\bar{g} + \underset{\bar{F}}{\underset{\parallel}{\bar{T}}} + 2\underset{\uparrow}{\bar{R}_V}$$

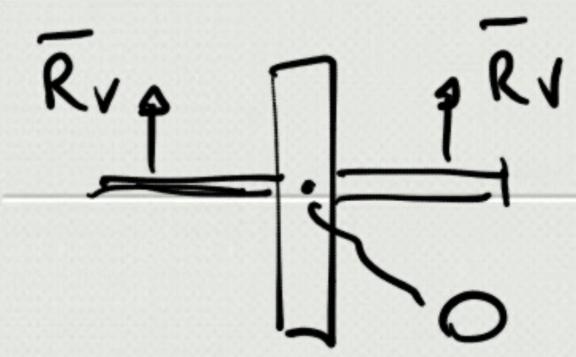


$\bar{M}^E_O = \bar{I}_z \bar{\alpha}$ $\qquad \bar{M} = \bar{r} \times \bar{F}$

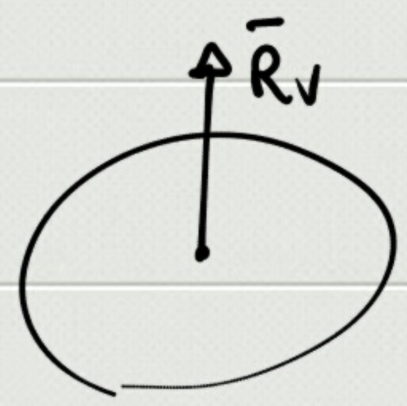
$\bar{M}_{peso,CM} = 0 \qquad (r = 0)$

$$\boxed{\bar{M}_T = \bar{r} \times \bar{T} = RT\bar{u}_z}$$





$\bar{M}_{R_V} = 0$

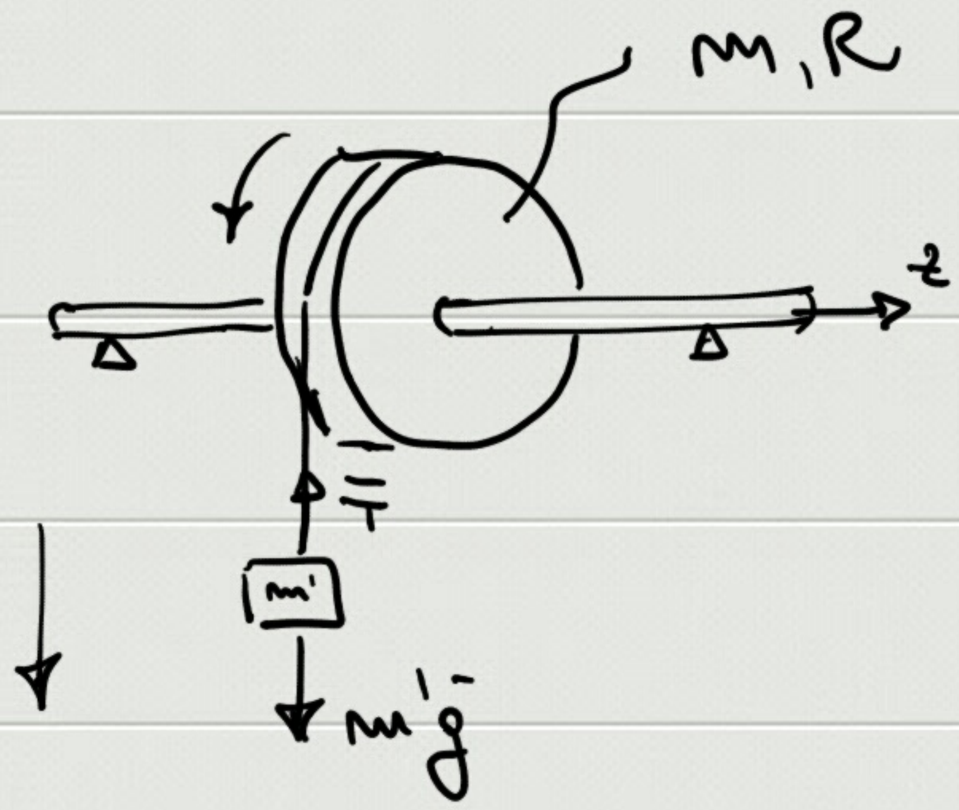


$M_{R_V} = 0$

$$\left.\begin{aligned} M^E_O &= RT \\ &\overset{!}{=} I_z \alpha = \frac{1}{2} m R^2 \alpha \end{aligned}\right\} \Rightarrow \alpha = \frac{2T}{mR} = \frac{2F}{mR} =$$

$$= 1.96 \text{ rad/s}^2$$

$\omega(t = 2s) = \alpha t = 3.92$ rad/s



$m' = 1\ kg$

$P = 9.8\ N$

$$M^E = \begin{cases} RT = I_z \alpha \\ m'g - T = m'a \\ a = a_T = \alpha R \end{cases}$$

$$\Rightarrow \cancel{R}T = \frac{1}{2} m R^{\cancel{2}} \alpha$$

$$m'g - T = m'\alpha R$$

$$m'g = \left(\frac{1}{2}m + m'\right)\alpha R$$

$$\Rightarrow \alpha = \frac{2m'g}{R(m + 2m')} = 1.8\ rad/s^2$$